Kevin Lu

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Education

Johns Hopkins University GPA: 3.93
Bachelor of Science in Computer Science and Neuroscience *Aug. 2022 – May 2026*

Experience

Senior Consultant and Full-Stack Engineer Nov. 2024 - Current
Emerald Consulting Group
Student-run company at JHU that provides business intelligence to biotech and life science firms *Baltimore, MD*

- Analyzed client KPIs and financial metrics (ROI, EBITDA, CAGR) to identify growth opportunities, delivering data-driven insights that informed strategic decision-making
- Developed market intelligence reports using industry benchmarks and TAM analysis, supporting clients' go-to-market strategies in the biotech and life sciences sector
- Developed Python-based data processing scripts for interactive analytics dashboards, integrating data extraction and transformation for business intelligence applications
- Conducted code reviews and performance optimizations, ensuring adherence to best coding practices and maintainability

Course Assistant for Data Structures Aug. 2024 – Current
Johns Hopkins University *Baltimore, MD*

- Held weekly office hours to explain challenging concepts with visual illustrations, debug students code and give tailored feedback on assignments

Machine Learning Engineer Aug. 2024 – Current
Johns Hopkins Medicine *Baltimore, MD*

- Trained and tested deep learning models for analyzing medical imaging and biosignal data, applying CNNs and transformers for classification and segmentation tasks
- Designed and deployed an end-to-end ML pipeline for anomaly detection in ECG signals, leveraging domain adaptation techniques to improve model robustness across datasets
- Optimized training infrastructure with GPU acceleration and distributed computing, reducing model training time by 40% and preventing process interruptions
- Synthesizing research findings into a primary-author manuscript, presented a poster session to 200+ clinicians at the annual *Human + AI: Redefining the Standard of Care in Medicine, 2025* .

Data Analyst May 2023 – Aug. 2023
University of Washington *Seattle, WA*

- Applied quantitative modeling and statistical analysis to deconstruct complex brainwave datasets, extracting actionable insights to support research recommendations
- Collaborated within cross-functional teams, leveraging structured frameworks and version control to ensure transparency, reproducibility, and alignment across deliverables

Projects

OrgoSolver: A Cloud-Based Organic Chemistry Web App | *React, Node, AWS* Aug. 2025 – Current

- Built a full-stack web app for solving Organic Chemistry synthesis problems, using AWS Lambda for serverless execution of synthesis algorithms and reaction pathway validation
- Developed a reinforcement learning algorithm to optimize synthetic reaction pathways, improving efficiency in multi-step synthesis planning
- Integrated S3 storage for handling molecule visualization assets, reducing server load and ensuring smooth rendering of complex chemical structures
- Deployed via AWS Amplify, streamlining development with automated builds and easy scaling

EquiHealth: A policy brief for physician owned hospitals Jan. 2025 – Current

- Conducted policy research on physician-owned hospitals, evaluating effects on healthcare access, cost efficiency, and competitive dynamics to inform strategic policy recommendations
- Collaborated with AANS/CNS's DC office and key stakeholders to draft a policy brief, integrating diverse perspectives into clear, evidence-based recommendations for legislative consideration